

Modeling Product Success in the Online Beauty Market: A Multi-Method Analysis of Amazon All Beauty Reviews



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Consumer Feedback & Product Success in Online Beauty

What signals in customer reviews predict whether a beauty product succeeds or fades on Amazon?

Research Questions

How do consumer feedback patterns and purchasing behaviors shape product trajectories in the online beauty market over time?

RQ1**NLP & Ratings**

Do star ratings alone capture satisfaction, or does sentiment reveal hidden dissatisfaction?

RQ2**Clustering**

Do products cluster into distinct segments by price tier and customer response?

RQ3**Time Series**

How have review volume and average ratings shifted over the 2015–2023 period?

RQ4**Association Rules**

Which early review characteristics co-occur most with long-term product success?

RQ5**Recommendation**

Can a hybrid model combining collaborative filtering and sentiment outperform a rating baseline?

Amazon All Beauty · Amazon Reviews 2023

701,528

Reviews

Nov 2000 – Sep 2023

112,590

Products

Metadata entries

21

Variables

After cleaning & merging

Key Variables

Variable	Source	Type	Description
rating	Review	Numeric 1–5	Star rating by reviewer
text	Review	Text	Free-text review body
helpful_vote	Review	Numeric	Helpfulness votes received
verified_purchase	Review	Binary	Purchase verified by Amazon
review_date	Review	DateTime	Date of submission
review_length	Derived	Numeric	Character count of review
price	Metadata	Numeric	Listed price in USD
price_tier	Derived	Ordinal	Budget / Mid / Premium / Luxury / Ultra
average_rating	Metadata	Numeric	Aggregate product rating
rating_number	Metadata	Numeric	Total ratings received

Cleaning Steps & 2015 Cutoff

Cleaning Steps

01

Deduplication

Removed 7,276 duplicate user-product review pairs — kept the most recent.

02

Timestamps

Converted Unix milliseconds to readable dates; extracted year and month.

03

Text Features

Derived character length, word count, and a flag for substantive reviews.

04

Price Handling

Price missing for 84% of products — built two separate working datasets.

05

Empty Fields

bought_together and categories were entirely empty — noted as limitations.

Why 2015?

Time series analysis showed that annual review counts stayed below twenty thousand before two thousand fifteen, compared to over thirty-five thousand from two thousand fifteen onward.

Pre-2015 data is too sparse for reliable trend estimation.

< 20K

reviews / year

Before 2015

> 35K

reviews / year

From 2015

Two Working Datasets

Full Review Set

All 2015+ reviews

NLP · Time Series · Recommendation

Priced Subset

Products with valid price

Clustering · Price-tier analysis

Do Star Ratings Capture True Satisfaction?



Fig. 6 — Star rating vs. review-text sentiment, 2015+ (sampled n=50,000)

Method

VADER scores cleaned review text (−1 to 1). Ratings are rescaled to the same range, so discrepancy = sentiment − scaled rating.

Findings

71.0% of reviews are 4–5 stars, but 3.8% of those (17,766 of 468,429) carry negative text sentiment — "hidden dissatisfaction."

Caution

VADER is lexicon-based and can misread sarcasm or mixed statements — a signal of dissatisfaction, not a diagnosis.

Does the Sentiment Gap Change as Products Mature?

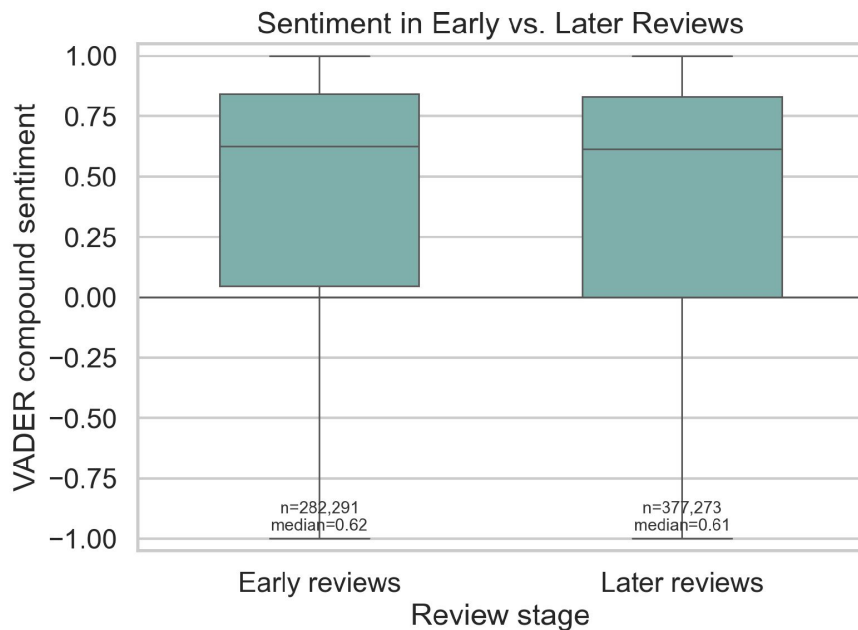


Fig. 7 — VADER compound sentiment, first 5 reviews vs. later reviews per product

Early vs. Later

Median sentiment holds steady: 0.62 for a product's first 5 reviews (n=282,291) vs. 0.61 afterward (n=377,273).

Dissatisfaction Creeps In

Hidden dissatisfaction (high rating, negative text) rises from 2.36% of early reviews to 2.94% of later ones.

Overall Distribution

Across all 659,564 reviews: 74.4% positive, 14.4% negative, 11.1% neutral text sentiment — reviews skew positive at every stage.

Review Volume & Rating Trends (2015–2023)

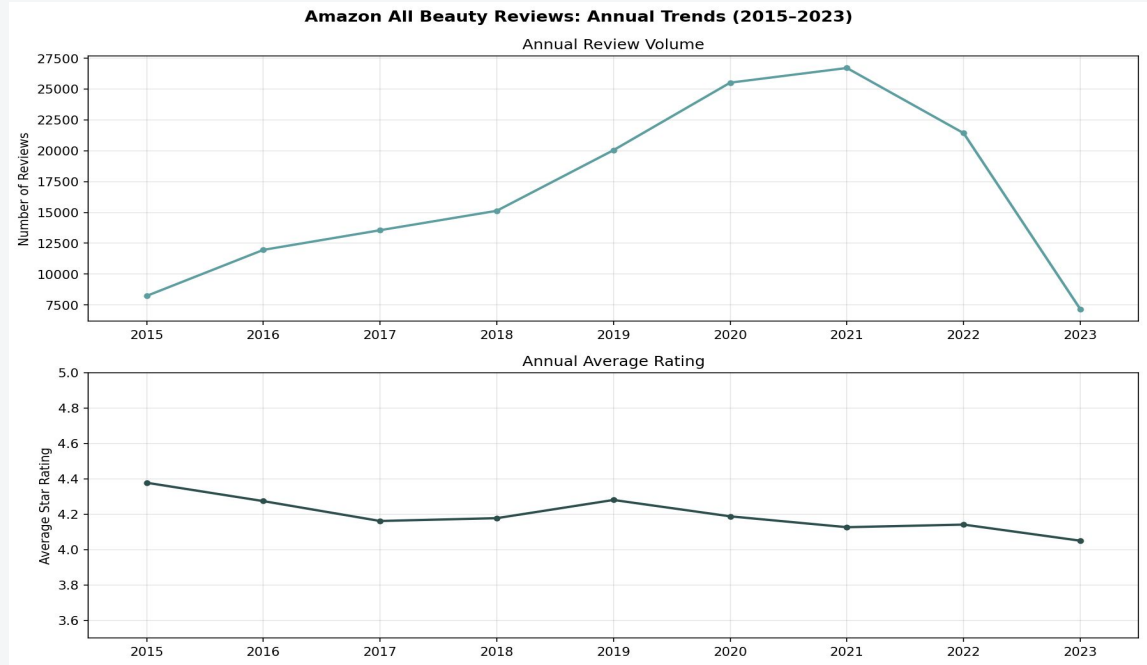


Fig. 1 — Annual review volume and average rating, 2000–2023



Volume

Steady growth from 2015 to a peak in 2021, followed by decline — reflecting data truncation at August 2023.



Ratings

Fell from 4.2 in 2013 to 3.7 in 2022 — growing consumer discernment as platform scaled.

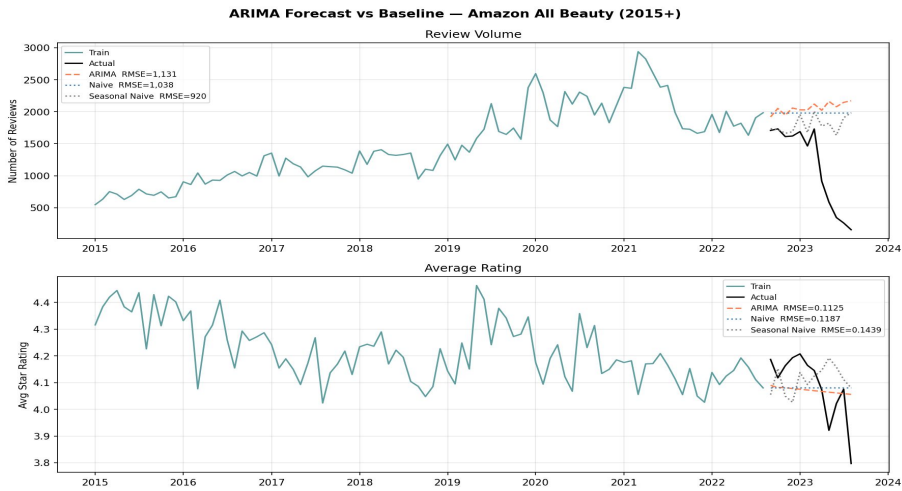


Seasonality

Mild volume patterns (± 500 /month). Ratings show virtually no seasonal movement (± 0.02).

ARIMA Forecasting & Cluster-Level Trends

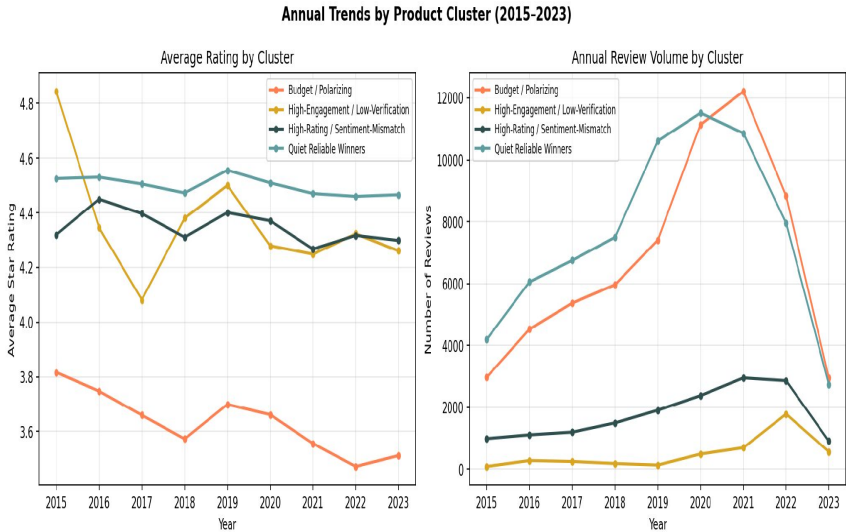
ARIMA Forecasting



Model Comparison — RMSE

Model	Volume RMSE	Rating RMSE
Naive	1,038	0.119
Seasonal Naive	920	0.144
ARIMA	1,131	0.113

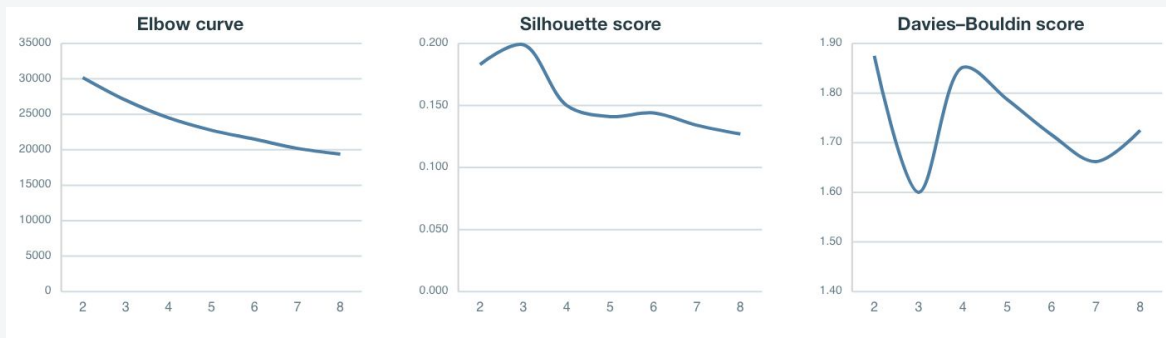
Trends by Product Cluster (2015–2023)



- Quiet Reliable Winners**
Stable ~4.5 throughout
- Budget / Polarizing**
Lowest ratings, declined post-2021
- High-Rating / Sentiment-Mismatch**
High rating but high variance
- High-Engagement / Low-Verification**
Smallest group, most volatile

Fig. 2 — ARIMA forecast vs baselines · Fig. 3 — Cluster trends 2015–2023

K = 4 adds an interpretable profile, despite weaker statistical fit



Analysis sample

5,996 products

Observed price and at least five reviews

13 standardized inputs

Price, ratings, verification, helpfulness, review detail and velocity, sentiment, mismatch, and early sentiment

Diagnostics favor K = 3

Highest silhouette (~0.199) and lowest Davies-Bouldin score.

K = 4 retained for interpretability

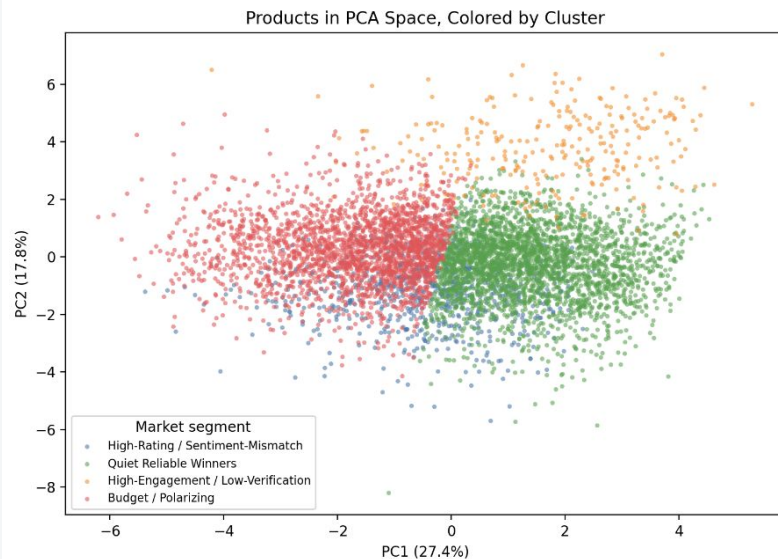
Its silhouette is ~0.150—not 0.174. K = 4 isolates the high-engagement, low-verification profile.

Conclusion

The four groups are overlapping working profiles, not uniquely optimal natural categories.

PCA shows response profiles that cut across every price tier

Amazon Beauty Product Segments (2015+ Real Review Sentiment)



PC1 + PC2 retain 45.2% of total variance.

All five price tiers appear within every cluster, so price influences—but does not determine—segment membership.

Four profiles differ most in response quality, mismatch, and engagement

Profile	Rating	Rating variation	Sentiment	Rating–text gap	Hidden dissatisfaction	Verified purchase	Reviews / month
Quiet Reliable Winners n=2,693	+0.7	-0.6	+0.6	-0.4	-0.3	+0.2	-0.1
Budget / Polarizing n=2,250	-1.0	+0.8	-0.7	+0.6	-0.2	+0.1	0.0
High-Rating / Sentiment-Mismatch n=785	+0.4	-0.2	-0.7	-0.4	+1.9	+0.2	-0.2
High-Engagement / Low-Verification n=268	+0.3	-0.5	+1.2	+0.2	-0.1	-3.8	+1.6

Cells show cluster means in standard-deviation units (z-scores)

Quiet Reliable Winners

Rating 4.54 · sentiment 0.57 · 96.2% verified

Budget / Polarizing

Rating 3.45 · sentiment 0.32 · highest rating variation

High-Rating / Sentiment-Mismatch

Rating 4.33 · sentiment 0.32 · 14.1% hidden dissatisfaction

High-Engagement / Low-Verification

2.55 reviews/month · 25.4% verified · n=268

PCA shows soft profiles—not rigid or causal categories

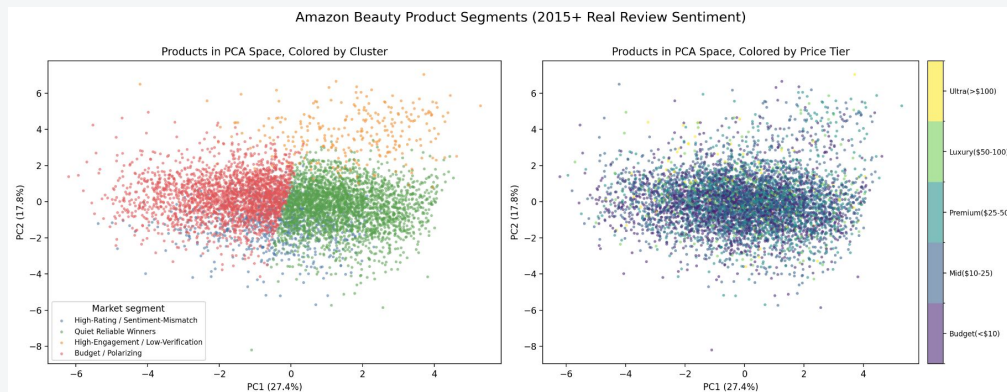


Fig. C4 · Cluster-colored PCA view (PC1 versus PC2)

How to interpret the map

- Nearby points have more similar standardized profiles.
- Overlap reflects continuous product behavior and modest separation.
- PC1 + PC2 retain 45.2% of total variance.
- PCA visualizes K-means output; it did not select K.

Decision-useful interpretation

Use these clusters for monitoring and strategy—not as rigid, causal, or fraud-detection categories.

Which Products Get Reviewed Together?

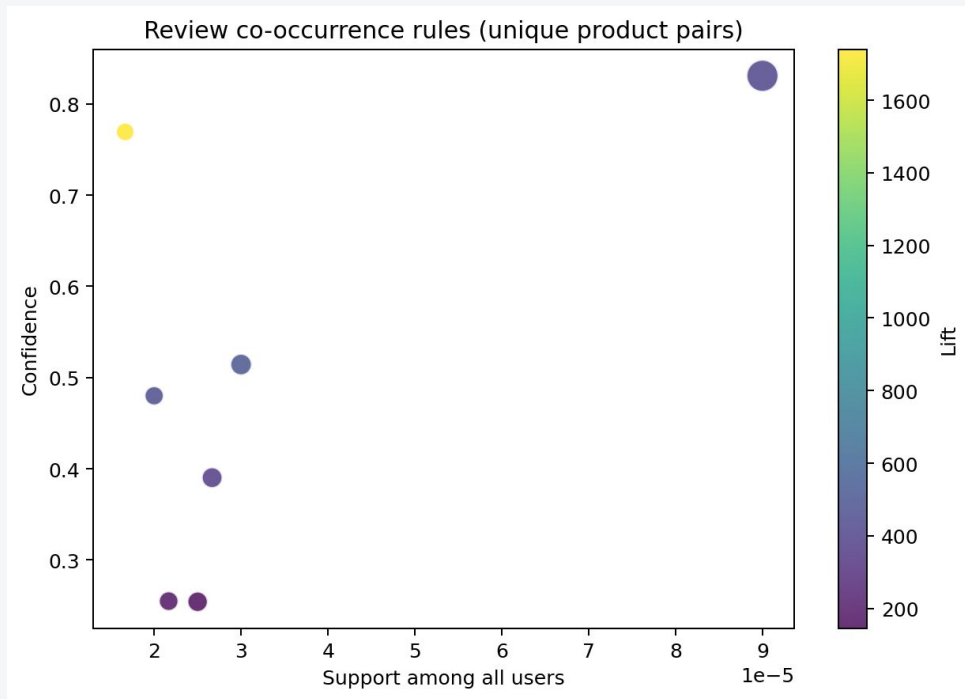


Fig. 4 — Review co-occurrence rules: support vs. confidence, colored and sized by lift



Method

bought_together is empty for every product. As a proxy, we treat products reviewed by the same user as a "basket," restricted to the 1,000 most-reviewed products for tractability.



Findings

7 rules survived (confidence $\geq 20\%$, lift ≥ 1.2), top lift $\sim 1,740$. Nearly all link variants of the same product line — beard oil scents, brush sets, hair-care lines — not surprising cross-category pairs.



Limitation

Review co-occurrence $\approx$ engagement, not confirmed co-purchase. Patterns apply only to the small, active slice of users who reviewed ≥ 2 products (6.8% of all reviewers).

Does Adding Sentiment Improve Recommendations?

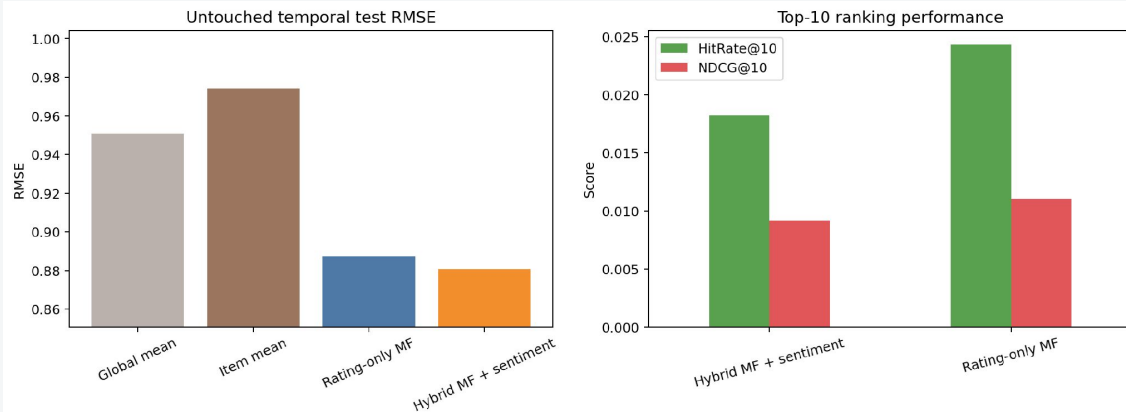


Fig. 5 — Untouched temporal test RMSE (left) and top-10 ranking performance (right)

806

active users evaluated (iterative k-core)

-0.75%

RMSE change, hybrid vs. rating-only

40

latent factors selected on validation

96.9%

test-set warm-start coverage



Method

Temporal leave-one-out split per user (most recent = test). Matrix factorization (SGD, bias terms) tuned on validation only; sentiment blended in and evaluated once on an untouched test set.



Results

RMSE: Rating-only 0.887 → Hybrid 0.881 (-0.75%). HitRate@10: Rating-only 2.43% → Hybrid 1.82%. NDCG@10: 0.0099 → 0.0092.



Verdict

Mixed evidence: sentiment sharpens rating predictions slightly but does not beat rating-only as a top-N recommender. Result is scoped to warm-start products and moderately active users (n=806).

Were the Research Questions Answered?

Research Question	Verdict	Evidence
RQ1 · NLP	Yes	71% of reviews are 4–5★, but 3.8% of those show negative text sentiment — star ratings alone miss real dissatisfaction.
RQ2 · Clustering	Yes	4 overlapping profiles emerge. Although K = 3 had the strongest statistical fit (silhouette ≈ 0.199), K = 4 was retained for interpretability (silhouette ≈ 0.150). The High-Engagement/Low-Verification profile contains 268 products and 25.4% verified purchases.
RQ3 · Time Series	Partial	ARIMA produced mixed results. For review volume, the seasonal-naive baseline performed best (RMSE 920), compared with the naive model (1,038) and ARIMA (1,131). For average ratings, ARIMA performed best (RMSE 0.113), narrowly outperforming the naive baseline (0.119).
RQ4 · Assoc. Rules	Limited	bought_together was empty for every product; the review-co-occurrence proxy surfaced only 7 rules among the 6.8% of multi-product reviewers.
RQ5 · Recommender	Mixed	Hybrid model cuts RMSE by 0.75%, but loses on Top-10 ranking (HitRate 1.82% vs. 2.43%) — no clean win either way.



Cross-Cutting Limitations

- 2015+ cutoff limits the “over time” story to 9 years, chosen for review-volume comparability, not the full 2000–2023 span.
- Every module leans on reviews as a stand-in for purchases — bought_together is empty, so co-purchase and true basket data are unavailable.
- Recommendation evaluation covers 806 active users (a real, but small and non-representative, warm-start core).
- VADER is lexicon-based and can misread sarcasm or mixed statements — hidden dissatisfaction is a signal, not a diagnosis.

Overall: across all five research questions, ratings alone are an incomplete signal. Text sentiment, engagement patterns, and co-review behavior each surface information ratings miss on their own — but every method here is scoped by the same data limitations: a 2015+ window, review-as-purchase-proxy, and small evaluation cores.

What Should Decision-Makers Do?



Audit high-rated reviews for hidden dissatisfaction

3.8% of 4–5-star reviews carry negative text sentiment. Route these to a lightweight QA queue — they're an early warning before it shows up in the aggregate rating.



Prioritize Cluster 2 for review-integrity review

268 products average 2.55 reviews per month but only 25.4% verified purchases (vs. 94.4–96.2% across the other profiles).



Bundle high-lift co-reviewed product pairs

Product-line variants reviewed together show lift up to ~1,740 (e.g., beard-oil scent variants, brush-set tiers) — natural candidates for cross-sell bundles or multipacks.



Hold the hybrid recommender before shipping it

Adding sentiment trims rating-prediction error by 0.75% but reduces Top-10 HitRate (1.82% vs. 2.43%). Keep the simpler rating-only model in production for now.

Expected Impact & Where to Go Next



Expected Impact

- Sentiment-vs-rating divergence can act as an early-warning signal — flagging dissatisfaction before it drags down the aggregate star rating.
- Auditing the low-verification, high-velocity segment (Cluster 2) protects the credibility of the platform's average ratings.
- Bundling the small set of high-lift co-reviewed product pairs targets a cross-sell opportunity that's cheap to test and already validated by real co-engagement.
- Keeping the simpler rating-only recommender avoids shipping a more complex hybrid model for a ranking-quality regression — saving engineering effort for a net loss.



Future Work & Potential Improvements

- Get true purchase / co-purchase data if Amazon exposes it — bought_together was empty, forcing every co-engagement measure here to rely on the review-as-proxy assumption.
- Extend the recommendation evaluation past the 806-user warm-start core as more interaction history accumulates.
- Try transformer-based sentiment models to catch sarcasm and mixed statements that lexicon-based VADER misses.
- Model time series per product cluster (as the professor suggested) to see whether trajectories genuinely differ by segment.
- Revisit the rating time-series model's differencing order — it currently underperforms a naive baseline, likely from over-differencing.